GRADIEND: Feature Learning within Neural Networks
Exemplified through Biases

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Overview and Method

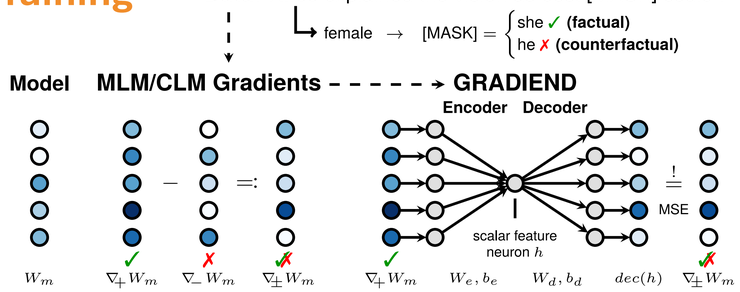
Summary

GRADIEND learns a single targeted **feature neuron** from model **gradients** and decodes it into a **persistent** model update that **changes** behavior with respect to that feature. We demonstrate this through social debiasing, focusing on gender and extending to race and religion.

What is GRADIEND?

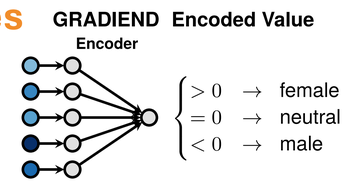
GRADIEND (GRADIent ENcoder Decoder) is an encoder-decoder method for targeted feature learning and controlled model rewriting based on **factual-counterfactual** gradients.

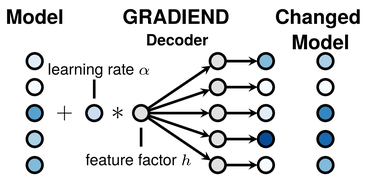
Training



Evaluation hypotheses

(H1) A targeted **feature** neuron with **desired** interpretation can be learned from model **gradients.**





(H2) This feature neuron enables **controlled modification** of feature-related model behavior without negatively affecting other capabilities.

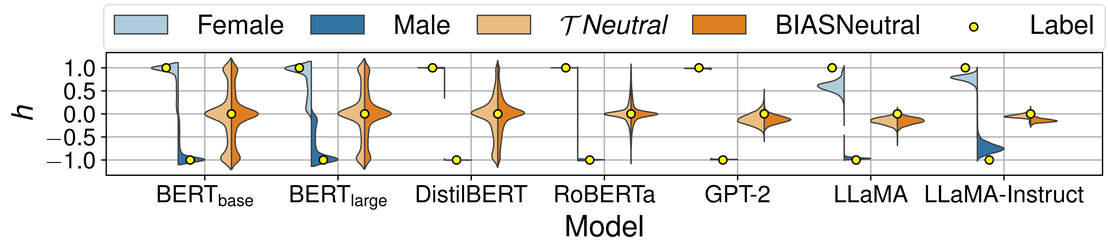
GRADIEND is a simple encoder-decoder

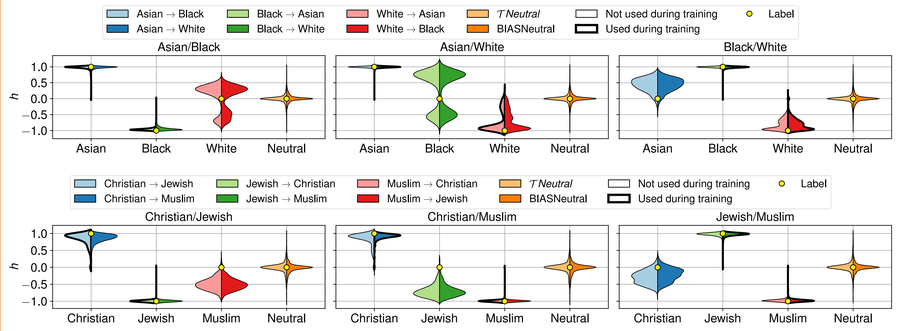
Objective: Learn $f(\nabla_+ W_m) \approx \nabla_\pm W_m$ with $f = dec \circ enc$
Encoder: $enc(\nabla_+ W_m) = \tanh(W_e^T \cdot \nabla_+ W_m + b_e) =: h$
Decoder: $dec(h) = h \cdot W_d + b_d$
Model update: $\widetilde{W}_m := W_m + \alpha \cdot dec(h)$

(H1) Targeted Feature Learning Works

The encoder separates feature classes

The encoder consistently **separates female** and **male** inputs across all evaluated models, while **neutral** inputs remain **distinct** from the feature classes.





Race and religion show that this works with non-binary features

GRADIEND is trained **pairwise** over the non-binary feature classes of race and religion. We show BERT-base encoded value distributions for readability; the same pattern holds across models.

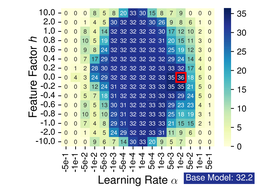
Neutral inputs remain distinct

Feature-unrelated targets stay apart from feature classes.

(H2) Controlled Model Rewrite Works

Main result: GRADIEND can be used for debiasing

Debiased models are selected on a search grid over feature factor h and learning rate α using the model update formula. Model selection jointly optimizes strong debiasing and good language modeling performance.



Variant			Prop. Rank Bias			Language Modeling (%)		
Name	ΔW	PP	Mean ↑	SS	SEAT	LMS	GLUE	SuperGLUE
GRADIEND + INLP	✓	✓	**0.88**	**0.91**	**0.84**	↓ -0.39 87.06	↓ -0.47 68.23	↓ -1.72 50.65
CDA + INLP	✓	✓	0.75	0.78	0.73	↑ 0.97 86.48	↑ 0.36 77.55	↑ 1.86 52.67
Dropout + INLP	✓	✓	0.71	0.78	0.64	↓ -1.09 84.42	↓ -2.43 74.75	↓ -0.80 50.01
INLP	✗	✓	0.67	0.62	0.72	↑ 0.10 87.56	↑ 0.13 68.83	↓ -0.82 51.55
GRADIEND + SentDebias	✓	✓	0.64	0.67	0.61	↓ -1.12 86.34	↓ -0.92 67.78	↓ -0.83 51.54
Dropout + SentDebias	✓	✓	0.62	0.70	0.55	↓ -3.25 82.27	↓ -2.25 74.93	↓ -0.21 50.60
SentDebias	✗	✓	0.60	0.48	0.72	↓ -0.52 86.94	↓ -0.44 68.27	↓ -0.08 52.29
CDA + SentDebias	✓	✓	0.57	0.71	0.43	↑ 0.01 85.52	↑ 0.50 77.68	↑ 1.25 52.06
GRADIEND	✓	✗	0.46	0.50	0.42	↓ -0.73 86.72	↓ -0.00 68.70	↓ -0.63 51.73
CDA	✓	✗	0.44	0.42	0.45	↑ 0.23 85.74	↑ 0.45 77.64	↑ 1.37 52.18
SelfDebias	✗	✓	0.41	0.41	–	↓ -9.65 77.81	–	–
LEACE	✗	✓	0.36	0.32	0.41	↓ -0.49 86.97	↑ 0.01 68.71	↓ -1.71 50.66
RLACE	✗	✓	0.31	0.21	0.40	↓ -2.19 85.26	↓ -0.06 68.64	↓ -1.85 50.51
Dropout	✓	✗	0.30	0.40	0.20	↓ -2.11 83.40	↓ -3.09 74.10	↓ -0.42 50.39
Base Model	✗	✗	0.17	0.11	0.23	87.46	68.70	52.37

Models are sorted by the Mean column. ΔW and PP indicate debiasing variant using weight modification and post-processing, respectively. Best variant type marked with a blue ✓.

GRADIEND is strongest weight-changing approach for gender debiasing

The table reports **gender bias metrics** StereoSet (SS) and SEAT, averaged across seven base models as **proportional rank**. Language modeling performance remains close to the base model.

Race and religion are harder to debias

GRADIEND also debiases race and religion bias, but less strongly than gender, likely due to noisier data, semantic ambiguity, and the restriction to a single rewrite axis.

Paper: https://iclr.cc/virtual/2026/poster/10011792
Code: https://github.com/aieng-lab/gradiend-bias

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